CHAPTER 4

DISTRIBUTION OF OCEANS AND CONTINENTS

CHAPTER ESSENCE AT A GLANCE:
Master the Continental Drift Theory of Alfred Wegener, paleomagnetic and fossil evidence, mantle convection currents, seafloor spreading by Harry Hess, and the modern synthesis of Plate Tectonics.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Continental Drift Hypothesis:** Alfred Wegener proposed in 1912 that all landmasses formed a singular supercontinent Pangaea surrounded by Panthalassa; fragmented into northern Laurasia and southern Gondwanaland during the Carboniferous period.

• **Empirical Lines of Evidence:** Shoreline matching (jigsaw fit of South America and Africa), geological age parity of trans-Atlantic orogenic belts, tillite glacial deposits across southern continents, and identical fossil fauna and flora (Mesosaurus, Glossopteris).

• **Thermal Mantle Convection and Seafloor Spreading:** Arthur Holmes postulated thermal convection currents in the mantle; Harry Hess synthesized this in 1961, proving new oceanic crust erupts at mid-ocean ridges and spreads symmetrically outward.

• **Plate Tectonics Paradigm:** Formulated by McKenzie, Parker, and Morgan (1967); lithosphere is fractured into 7 major and numerous minor rigid plates interacting along Divergent (constructive), Convergent (destructive), and Transform (conservative) boundaries.

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$\text{Seafloor Spreading Rate } (v) = \frac{\Delta \text{Distance}}{\Delta \text{Geological Time}}$$

Kinematic Rate of Oceanic Crust Creation at Mid-Ocean Ridges

$$\text{Thermal Mantle Convection} \rightarrow \{\text{Ascending Plume} \rightarrow \text{Ridge}, \quad \text{Sinking Slab} \rightarrow \text{Trench}\}$$

Holmes Convection Dynamic Governing Plate Drift

$$\text{Plate Boundary Kinematics} \rightarrow \text{Divergent (Constructive)} + \text{Convergent (Destructive)} + \text{Transform}$$

Tripartite Structural Classification of Lithospheric Boundaries

$$\Delta B(x) = B_{\text{measured}} - B_{\text{regional}} \rightarrow \text{Symmetrical Paleomagnetic Stripping}$$

Vine-Matthews-Morley Magnetic Reversal Anomaly Metric

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Wegener's suggested driving forces (pole-fleeing force due to Earth's rotation and tidal force from the Moon and Sun) were scientifically rejected as inadequate; thermal mantle convection is the verified mechanism.

• **Alert:** Oceanic crust is geologically young everywhere (under 200 million years old) compared to ancient continental rocks (up to 3.8 billion years old) because oceanic crust continuously subducts into deep trenches and recycles into the mantle.

• **Alert:** Transform boundaries do not create or destroy crust; plates slide horizontally past one another along transform faults, giving rise to shallow-focus earthquakes (e.g. San Andreas Fault).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What were Pangaea and Panthalassa according to Alfred Wegener's Continental Drift Theory? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Pangaea:** Wegener postulated that around 250 million years ago, all continents were united as a single supercontinent named Pangaea, meaning 'all Earth'. **[1 Mark]**

• **Panthalassa:** The single mega-ocean that surrounded the supercontinent Pangaea, meaning 'all water'. **[1 Mark]**

2. Name any two driving forces proposed by Alfred Wegener for continental drift and state why they were rejected. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Two Forces:** Pole-fleeing force (caused by Earth's centrifugal rotation) and Tidal force (caused by gravitational attraction of the Sun and Moon). **[1 Mark]**

• **Reason for Rejection:** Physicists proved that tidal and rotational forces are millions of times too weak to break continents and move rigid continental masses across ocean basins. **[1 Mark]**

3. Explain three important geological and fossil evidences supporting the Continental Drift Theory. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **1. Jigsaw Fit of Coastlines:** The opposing coastlines of the Atlantic Ocean, particularly South America and Africa, display a remarkable complementary fit at the 1000-fathom contour. **[1 Mark]**

• **2. Rocks of Same Age Across Oceans:** A belt of ancient marine rocks aged 2,000 million years from Brazil coast matches identical rock systems along the coastline of western Africa. **[1 Mark]**

• **3. Fossil Distribution:** Identical fossils of the small freshwater reptile Mesosaurus are found exclusively in southern Africa and eastern Brazil, species incapable of swimming across the ocean. **[1 Mark]**

4. What is Seafloor Spreading? Explain the observations that led Harry Hess to propose this concept. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Concept:** Proposed by Harry Hess in 1961; constant volcanic eruptions along mid-ocean ridges rupture oceanic crust and push newly formed basaltic crust laterally outwards away from the ridge axis. **[1.5 Marks]**

• **Supporting Observations:** Rocks equidistant from mid-ocean ridge crests show identical chemical composition, age, and magnetic polarity stripes, and oceanic crust is nowhere older than 200 million years. **[1.5 Marks]**

5. Describe the three types of plate boundaries (Divergent, Convergent, Transform) with examples and associated geological features. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Divergent Boundaries (Constructive):** Where plates pull apart from each other and new crust is formed by ascending magma (e.g. Mid-Atlantic Ridge, East African Rift). **[1.5 Marks]**

• **Convergent Boundaries (Destructive):** Where plates collide and one plate slides under another into the mantle at subduction zones, forming deep trenches, volcanic arcs, and fold mountains (e.g. Himalayas, Mariana Trench). **[1.5 Marks]**

• **Transform Boundaries (Conservative):** Where plates slide past each other horizontally along transform faults without creating or destroying crust (e.g. San Andreas Fault in California). **[1 Mark]**

6. Name the seven major lithospheric plates and explain the driving mechanism that causes them to move.
[Score: 4 Marks]

Answer & Key Valuation Steps:

• **Seven Major Plates:** (1) Antarctica and surrounding ocean plate, (2) North American plate, (3) South American plate, (4) Pacific plate, (5) India-Australia-New Zealand plate, (6) African plate, and (7) Eurasian plate. **[2 Marks]**

• **Driving Mechanism:** Driven by thermal convection cells operating within the asthenosphere and mantle, generated by radioactive decay and residual core heat. Heated magma ascends at ridges and cool dense lithospheric slabs sink at subduction zones. **[2 Marks]**